

03 – C

Let's C

Jérôme Landré – jerome.landre@ju.se

OUTLINE

- 1) History
- 2) Small example to start with
- 3) Live coding examples

1) HISTORY

- The **C** programming language was **developed in the early 1970s by Dennis Ritchie at Bell Labs**. Ritchie's goal was to create a portable, efficient, and flexible language that could be used for systems programming
 - **1972: The First Version** Dennis Ritchie and Brian Kernighan, another Bell Labs researcher, collaborated on the first version of C, which was called "K&R C" (after their surnames). This initial version was designed to be a minimalist language, focusing on simplicity and efficiency.
 - **1978: Publication of K&R Book** Kernighan and Ritchie published their book, "The C Programming Language," which became a seminal work in the field. The book introduced the language to a wider audience and established many of the conventions and standards that are still followed today.
 - **1983: ANSI Standardization** The American National Standards Institute (ANSI) standardized C as ANSI C, which was published in 1983. This standardization effort helped to solidify C's position as a widely adopted and widely supported language.

1) HISTORY

- **1990s: Evolution and Extensions** In the 1990s, C evolved with the introduction of new features and extensions, such as:
 - **C89**: The 1989 standard added support for functions, arrays, and structures.
 - **C90**: The 1990 standard refined the language and added support for more advanced features.
 - **C99**: The 1999 standard introduced features like variable-length arrays, designated initializers, and improved support for internationalization.
- **Modern C**
 - Today, C remains a popular and influential language, widely used in operating systems, embedded systems, and other applications. The previous standard was ISO/IEC 9899:2018 also known as C17 (**AI hallucination** here in 2024: it was mentioned **C11** by Brave browser AI) The current standard is ISO/IEC 9899:2024, also known as **C23**. Clang, a modern compiler, continues to advance the language, providing features like static analysis and improved error messages.

Source: Brave browser AI, it did an error on **C17**, it was named **C11**, [https://en.wikipedia.org/wiki/C17_\(C_standard_revision\)](https://en.wikipedia.org/wiki/C17_(C_standard_revision))

1) HISTORY

- **Key Features** Throughout its history, C has maintained its focus on:
 - Low-level memory access
 - Efficiency and portability
 - Simplicity and readability
 - Modular framework
- These features have contributed to C's enduring popularity and versatility, making it a fundamental language in the world of computer science.

2) SMALL EXAMPLE TO START WITH

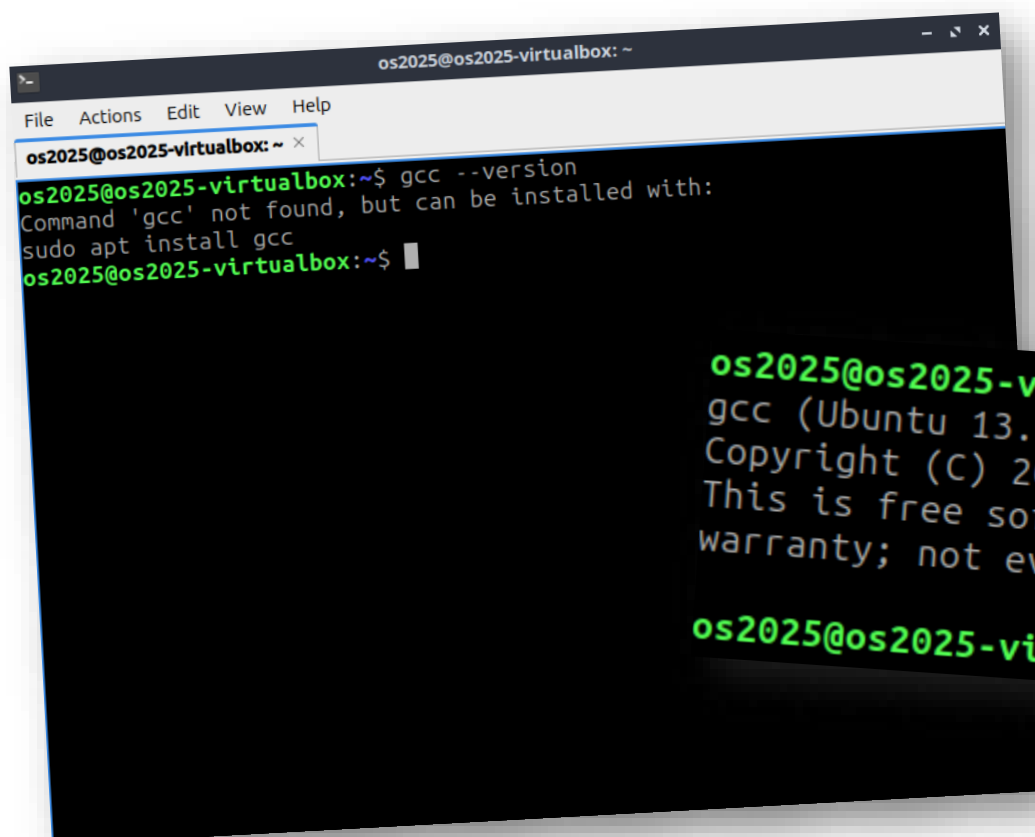
- Before **starting**: Do I have gcc installed?

```
sudo apt update
```

```
sudo apt upgrade
```

```
sudo apt autoremove
```

```
sudo apt install build-essential
```



```
os2025@os2025-virtualbox: ~  
File Actions Edit View Help  
os2025@os2025-virtualbox: ~  
os2025@os2025-virtualbox:~$ gcc --version  
Command 'gcc' not found, but can be installed with:  
sudo apt install gcc  
os2025@os2025-virtualbox:~$
```

```
os2025@os2025-virtualbox:~$ gcc --version  
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0  
Copyright (C) 2023 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
os2025@os2025-virtualbox:~$
```

2) SMALL EXAMPLE TO START WITH

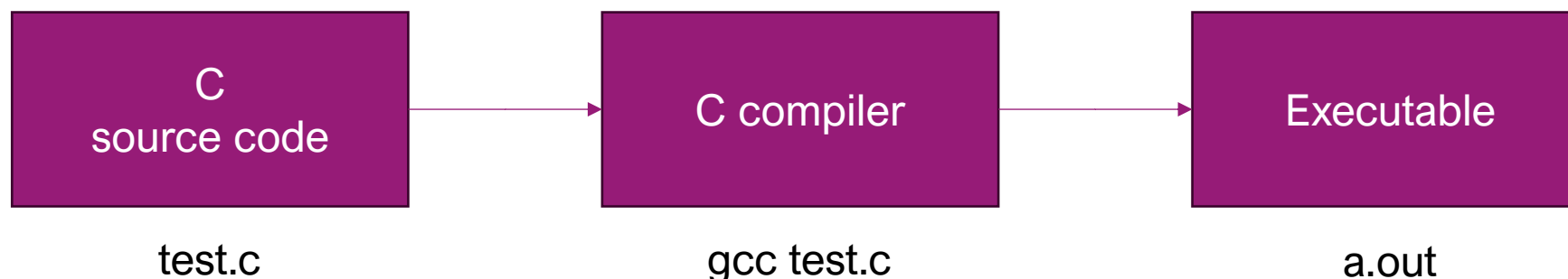
- Example from ChatGPT, which is very good at coding! (but try and check the given code by yourself)
- This is the **source** code, we must **compile** it to get an **executable** file.

```
#include <stdio.h>

int main() {
    // Print Hello, World! to the console
    printf("Hello, World!\n");

    return 0; // Return success code
}
```

2) SMALL EXAMPLE TO START WITH



```

os2024@os2024-virtualbox:~$ pwd
/home/os2024
os2024@os2024-virtualbox:~$ mkdir c-code
os2024@os2024-virtualbox:~$ cd c-code
os2024@os2024-virtualbox:~/c-code$ ls -la
total 8
drwxrwxr-x  2 os2024 os2024 4096 okt 21 09:32 .
drwxr-x--- 16 os2024 os2024 4096 okt 21 09:32 ..
  
```

```

os2024@os2024-virtualbox:~/c-code$ gcc test.c
os2024@os2024-virtualbox:~/c-code$ ls -la
total 28
drwxrwxr-x  2 os2024 os2024 4096 okt 21 09:35 .
drwxr-x--- 16 os2024 os2024 4096 okt 21 09:32 ..
-rwxrwxr-x  1 os2024 os2024 15952 okt 21 09:35 a.out
-rw-rw-r--  1 os2024 os2024   73 okt 21 09:35 test.c
  
```

```

os2024@os2024-virtualbox:~/c-code$ vi test.c
  
```

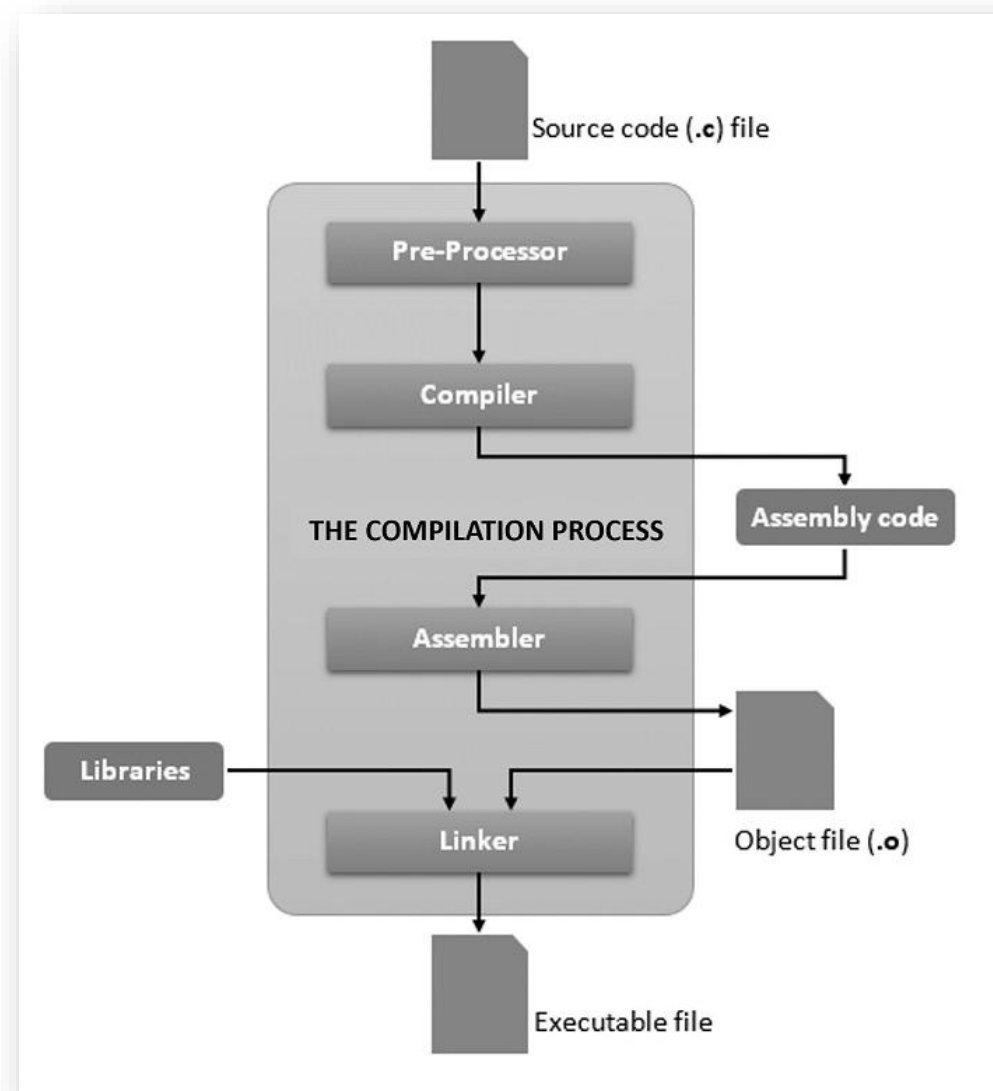
```

1 #include <stdio.h>
2
3 int main() {
4     printf("Hello, World!");
5     return 0;
6 }
7
  
```

```

os2024@os2024-virtualbox:~/c-code$ ./a.out
Hello, World!os2024@os2024-virtualbox:~/c-code$
  
```


2) SMALL EXAMPLE TO START WITH



3) LIVE CODING EXAMPLES

LIVE CODING...

3) LIVE CODING EXAMPLES

```
1 // This line includes the standard inputoutput library to use printf()
2 #include <stdio.h>
3 /*
4  * This is the main function
5  * */
6 int main() {
7     // printing the Hello, World! text
8     printf("Hello, World!\n");
9     // returning a value (integer)
10    return 0;
11 }
12
```

ANY QUESTIONS?